Problem 2:-

A box contains three coins: two regular coins and one fake two-headed coin
(P(H)=1P(H)=1),
• You pick a coin at random and toss it. What is the probability that it lands heads up?
• You pick a coin at random and toss it, and get heads. What is the probability that it is the
two-headed coin?

Let's solve this problem step by step using the concept of **conditional probability** and **Bayes' theorem**.

Problem 2

A box contains **3 coins**:

- Coin 1 = Regular coin (H or T are equally likely)
- Coin 2 = Regular coin (H or T are equally likely)
- Coin 3 = Two-headed coin (always lands Heads)

Each coin is equally likely to be selected.

Part (a): Probability that the toss results in Heads

Step 1: Probability of selecting each coin

Since there are three coins,

$$P(\text{Regular Coin 1}) = \frac{1}{3}$$

$$P(\text{Regular Coin 2}) = \frac{1}{3}$$

$$P(\text{Two-headed Coin}) = \frac{1}{3}$$

Step 2: Probability of getting Heads from each coin

- Regular coin:

$$P(H|\text{Regular}) = \frac{1}{2}$$

- Two-headed coin:

$$P(H|\text{Two-headed}) = 1$$

Step 3: Apply the Law of Total Probability

$$P(H) = P(H|R_1)P(R_1) + P(H|R_2)P(R_2) + P(H|T)P(T)$$

Substitute the values:

$$P(H) = \frac{1}{2}\left(\frac{1}{3}\right) + \frac{1}{2}\left(\frac{1}{3}\right) + 1\left(\frac{1}{3}\right)$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{3}$$

Convert to a common denominator:

$$= \frac{1}{6} + \frac{1}{6} + \frac{2}{6}$$

$$= \frac{4}{6}$$

$$= \boxed{\frac{2}{3}}$$

Answer (a):

$$\boxed{P(H) = \frac{2}{3}}$$

Part (b): Given that the toss is Heads, what is the probability that the selected coin is the two-headed coin?

We need

$$P(\text{Two-headed Coin}\,|H)$$

Use **Bayes' Theorem**:

$$P(T|H) = \frac{P(H|T)P(T)}{P(H)}$$

Where

- $P(H|T) = 1$
- $P(T) = \frac{1}{3}$
- $P(H) = \frac{2}{3}$

Substitute:

$$P(T|H) = \frac{1 \times \frac{1}{3}}{\frac{2}{3}}$$

Multiply by the reciprocal:

$$= \frac{1}{3} \times \frac{3}{2}$$

$$= \frac{1}{2}$$

Answer (b):

$$\boxed{P(\text{Two-headed Coin } | H) = \frac{1}{2}}$$

Alternative Explanation (Using a Table)

Selected Coin	Probability of Selecting	Probability of Heads	Joint Probability
Regular Coin 1	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$
Regular Coin 2	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$
Two-headed Coin	$\frac{1}{3}$	1	$\frac{1}{3}$

Total probability of getting Heads:

$$\frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3}$$

Among the Head outcomes, the contribution from the two-headed coin is:

$$\frac{1}{3}$$

Therefore,

$$P(\text{Two-headed Coin}|H) = \frac{\frac{1}{3}}{\frac{2}{3}} = \boxed{\frac{1}{2}}$$

Final Answers

1. **Probability that the coin lands Heads:**

$$\boxed{\frac{2}{3}}$$

2. **Probability that the selected coin is the two-headed coin, given that the toss resulted in Heads:**

$$\boxed{\frac{1}{2}}$$

Interpretation: Even after observing a head, there is only a **50% chance** that you selected the two-headed coin, because a regular coin can also produce a head with probability $\frac{1}{2}$.



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